

1. Alice rolls a 6-sided dice, and Bob rolls an 8-sided dice. Let X denote the sum of the two values. Let X_1 denote Alice's roll, and let X_2 denote Bob's roll, so $X = X_1 + X_2$, and the two X_j s are independent. Find $\text{Var}(X)$ by first finding $\text{Var}(X_1)$ and $\text{Var}(X_2)$.

2. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six equally likely colors, one of which is green. Let X denote the number of green bears in a randomly selected bear family.

2a. Find $E(X^2)$ using the probability mass function of X , as in Problem Set 7, question 3.

2b. Find $E(X^2)$ in a different way, as follows: Let $X_1 = 1$ if the daddy bear is green, and $X_1 = 0$ otherwise. Let $X_2 = 1$ if the momma bear is green, and $X_2 = 0$ otherwise. Let $X_3 = 1$ if the baby bear is green, and $X_3 = 0$ otherwise. Now $X = X_1 + X_2 + X_3$, where the X_j s are independent. Expand $(X_1 + X_2 + X_3)^2$ into 9 terms, where 6 of them will behave one way, and the other 3 will behave another way.

2c. Find $\text{Var}(X)$ using $E(X^2)$ and your answer to $E(X)$ (from Problem Set 10 or 11).

2d. Find $\text{Var}(X)$ in a different way, namely, using $\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3)$, since the X_j s are independent in this problem.

3. Suppose a drawer of t-shirts contains 7 t-shirts from Purdue and 3 t-shirts from IU. Alice randomly selects t-shirts from the drawer, **without replacement**, until the first IU t-shirt is selected. All possible orders of selection are equally likely. Let X denote the number of t-shirts she chooses, until the first IU t-shirt appears (including the IU t-shirt itself). Label the 7 Purdue t-shirts with the numbers 1 to 7, to keep track of which is which.

Let $X_j = 1$ if the j th labelled Purdue shirt is chosen before the first IU t-shirt (and therefore before all of the IU t-shirts), and $X_j = 0$ otherwise. Notice that $X = 1 + X_1 + X_2 + \cdots + X_7$, where the "1" is because we always need 1 shirt for the IU t-shirt itself.

3a. Find $E(X^2)$ using the probability mass function of X , as in Problem Set 7, question 4. (Hint: You might want to first normalize the pmf to have 120 in each denominator.)

3b. Find $E(X^2)$ in a different way: Expand $(1 + X_1 + X_2 + \cdots + X_7)^2$ into 64 terms.

3c. Find $\text{Var}(X)$ using $E(X^2)$ and your answer to $E(X)$ (from Problem Set 10 or 11).

3d. Are the X_j s independent? Why or why not?

4. During a 5-day week (Monday to Friday), each day is sunny, cloudy, or rainy; these days' weather types are independent and equally likely. Let X denote the number of sunny days during a 5-day weather prediction. Let $X_j = 1$ if the j th day is predicted to be sunny, and $X_j = 0$ otherwise. Notice that $X = X_1 + \cdots + X_5$.

4a. Find $E(X^2)$ using the probability mass function of X , as in Problem Set 2, question 4.

4b. Find $E(X^2)$ in a different way, as follows: Expand $(X_1 + \cdots + X_5)^2$ into 25 terms, where 20 of them will behave one way, and the other 5 will behave another way.

4c. Find $\text{Var}(X)$ using $E(X^2)$ and your answer to $E(X)$ (from Problem Set 10 or 11).

4d. Find $\text{Var}(X)$ in a different way, namely, using $\text{Var}(X) = \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_5)$, since the X_j s are independent in this problem.

5. Consider X with $P(X = x) = \frac{3^x e^{-3}}{x!}$ for integers $x \geq 0$.

5a. Write $x^2 = (x)(x-1) + x$ and then compute $E(X^2)$.

5b. Using $E(X)$ from Problem Set 10, Question 5, and $E(X^2)$ from 5a, find $\text{Var}(X)$.